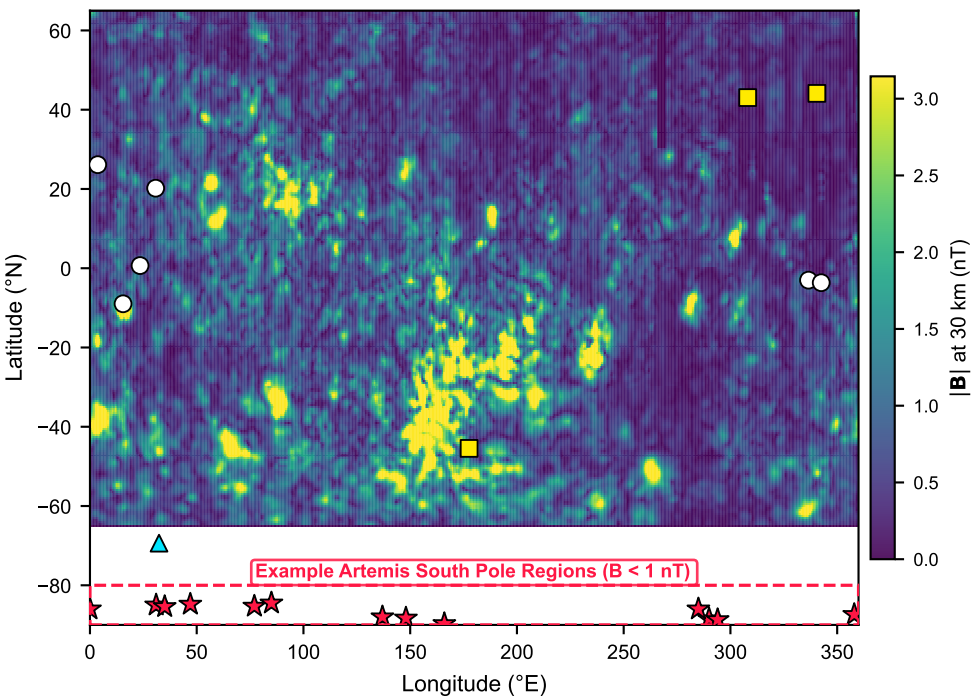
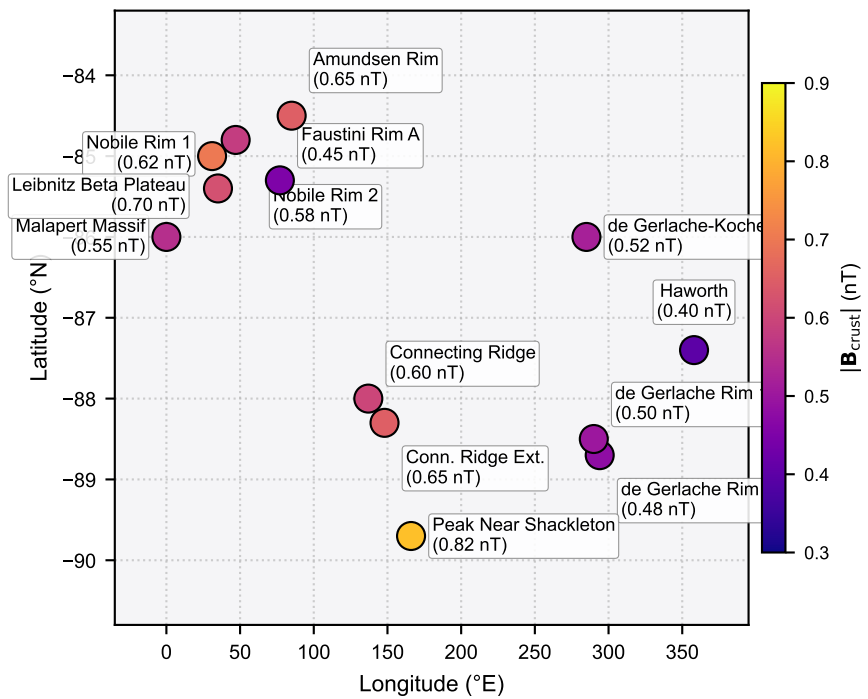
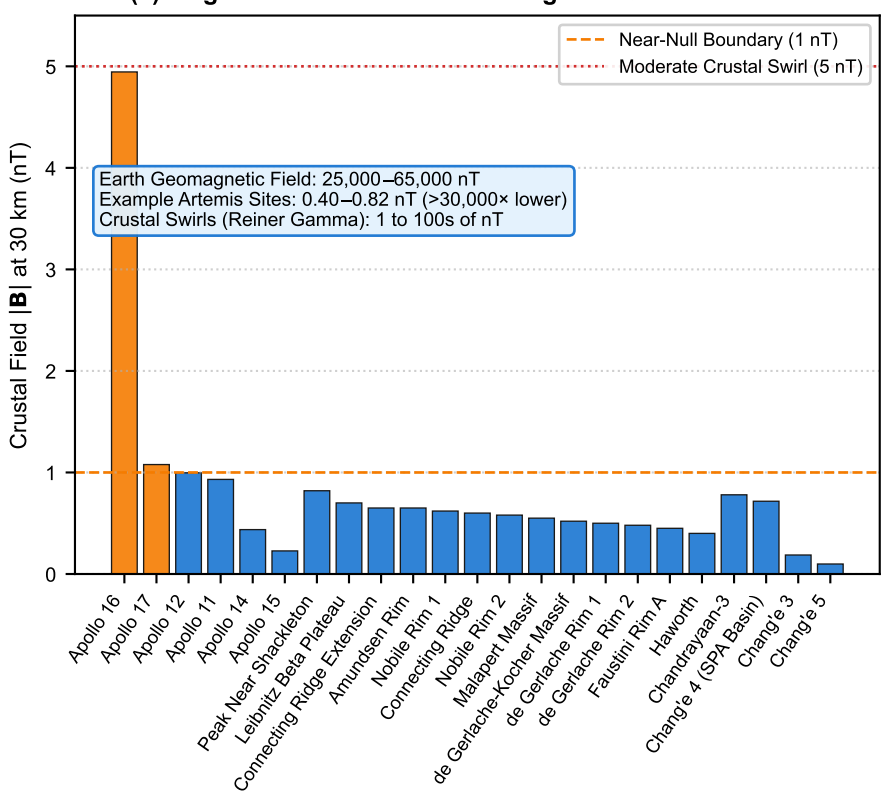






(d) Field Superposition at the Biological Volume

$$\mathbf{B}_{\text{total}}(t, \mathbf{x}) = \mathbf{B}_{\text{crust}} + \mathbf{B}_{\text{ext}} + \mathbf{B}_{\text{lander}} + \mathbf{B}_{\text{payload}} + \mathbf{B}_{\text{control}}$$

Net Vector Sum Governs Primary Spin-Chemical Biomarkers

Lunar Plant Growth Enclosure (e.g., LEAF / BLSS)

$\mathbf{B}_{\text{crust}}$	**[0.4 – 0.8 nT]**	Lunar South Pole paleomagnetic void
$\mathbf{B}_{\text{ext}}$	**[1 – 5 nT]**	Solar wind IMF / magnetotail shadow
$\mathbf{B}_{\text{lander}}$	**[50 – 500 µT]**	Lander avionics, DC power buses & batteries [Confounder]
$\mathbf{B}_{\text{payload}}$	**[15 – 20 nT]**	Payload fans, fluidic pumps, heaters, LEDs [Ripple]
$\mathbf{B}_{\text{control}}$	**[40 – 50 µT]**	Active 3-axis compensation / causal rescue coils

CRITICAL: Unshielded spacecraft DC currents (50–500 µT) can exceed Earth GMF, inadvertently masking true lunar hypomagnetic exposure!